

VIMAL KUMAR R N

Senior Hardware Engineer | Hardware Product Manager

Connected Wearables • Safety-Critical IoT • EU + US Certified Products • 50K+ units/year shipped

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Featured on Shark Tank India S1 • All-India Winner, SAE-India BAJA 2018

PROFESSIONAL SUMMARY

Hardware Engineer and Associate Product Manager with 7+ years owning connected-device development end-to-end — from schematic and BLE / sensor-fusion architecture through DFMEA, China-vendor production, and FCC / CE / RED EN 18031 certification. Currently lead hardware and product for Quin's crash-detection safety platform shipping 50,000+ units annually across US, EU, and emerging markets. Co-founded a medical wearable startup (₹1.27 Cr funded). Equally comfortable owning the schematic and the PRD.

CORE COMPETENCIES

Hardware Design: Analog and digital circuit design, mixed-signal systems, high-speed multi-layer PCB (up to 8 layers), signal integrity, power integrity, EMC, thermal design, low-power optimization

PCB Tools: Altium Designer, Cadence Allegro, Mentor Graphics, KiCAD, EasyEDA, Eagle

Wireless & Connectivity: BLE 5.0, WiFi 6, GSM / LTE, GPS, NFC, RFID, LoRa, Zigbee

Communication Protocols: SPI, I2C, UART, CAN, LIN, USB, Ethernet

Microcontrollers: ARM Cortex-M (M0 / M3 / M4 / M7), STM32, ESP32, Nordic nRF52, TI MSP430

Sensors & Systems: IMU, accelerometer, gyroscope, proximity, bio-signal, environmental sensors, sensor fusion, event classification

Compliance & Standards: FCC, CE, RED EN 18031, ISO 27001, ISO 26262, AUTOSAR, GDPR, RoHS, IPC-2221, medical device standards

Production & NPI: DFM, DFMEA, EVT / DVT / PVT, RCCA, yield optimization, high-volume production (50K+ units/year)

Supply Chain: China vendor sourcing, supplier qualification, BOM cost optimization, contract negotiation

Product Management: PRD authoring, roadmap planning, Agile, Jira, Confluence, stakeholder management, GTM, voice of customer

Test & Validation: Oscilloscopes, logic analyzers, spectrum analyzers, NI DAQ, thermal imaging, LTspice, MATLAB

PROFESSIONAL EXPERIENCE

Senior Hardware Engineer & Associate Product Manager

August 2022 – Present

Quin Design, Hyderabad

- Architect crash-detection and safety hardware for connected helmets — IMU, 3-axis accelerometer, gyroscope, proximity, BLE 5.0, GSM / LTE, GPS — delivering sub-10ms event response, 10+ days standby, IP67 protection in a wearable form factor.
- Own product roadmap and PRDs across 6+ industrial electronics programs spanning hardware, firmware, mobile app, and backend cloud teams; consistent spec acceptance from leadership and clients.
- Lead PCB design reviews and layout governance focused on first-pass success, signal and power integrity, EMC robustness, and manufacturability; support production scaling to 50,000+ units shipped annually.
- Drove meaningful BOM cost reduction and supplier cost savings via component re-engineering, alternate sourcing, and direct China-vendor negotiation across 5+ key suppliers.
- Spearheaded global certification strategy across FCC, CE, and RED EN 18031; authored technical files, risk assessments, and GDPR-compliant cloud architecture with EU data residency on AWS.
- Implemented DFMEA framework and led RCCA investigations on field returns; drove corrective actions that reduced recurring defect modes and improved field reliability.
- Recruited and mentored hardware engineering team; authored cross-functional PRDs enabling delivery with mobile, backend, and analytics teams; delivered to 30+ global clients across US, EU, and emerging markets.

Co-Founder & CTO

December 2020 – July 2022

VivaLyf Innovations Pvt. Ltd., Hyderabad

- Co-founded medical wearable startup; raised ₹1.27 Cr (~\$127K) seed funding; featured on Shark Tank India Season 1.

- Designed 8-layer mixed-signal biosensor system with >85 dB SNR in a compact wearable form factor; achieved 72-hour battery life through power-optimization techniques.
- Owned full product lifecycle — concept, system architecture, hardware design, firmware coordination, EVT / DVT, regulatory pathway — and led clinical partner engagement with 10+ healthcare institutions.
- Built and led cross-functional team across hardware, firmware, algorithms, and QA under startup resource constraints; managed investor and clinical-partner communication.

Program Associate — Hardware Innovation

May 2019 – August 2020

Forge Accelerator, Coimbatore

- Directed hardware development and rapid prototyping for 50+ early-stage startups across industrial automation, consumer electronics, and connected devices.
- Orchestrated open-innovation programs with 50+ startups and 20 corporate partners; converted 10 pilots into 3 commercial partnerships.
- Built reference designs integrating WiFi, Bluetooth, and LoRa with SPI / I2C / UART / CAN; established rapid-prototyping framework that accelerated concept-to-prototype delivery.

Associate Software Engineer — Automotive Systems

September 2018 – May 2019

Robert Bosch Engineering and Business Solutions, Coimbatore

- Developed instrument-cluster software for Volkswagen Group and BMW under ISO 26262 and AUTOSAR; collaborated across software, hardware, and validation teams.
- Built the Navonmesha gadget library covering 1000+ automotive-grade components, accelerating concept-to-prototype across teams; presented at 3 industry conferences.

EDUCATION

M.Tech, Embedded Systems | 2024 – 2026

Birla Institute of Technology and Science (BITS) Pilani — Work Integrated Learning Programme

B.E., Electrical and Electronics Engineering | 2014 – 2018

Government College of Technology, Anna University, Coimbatore

AWARDS, CERTIFICATIONS & RECOGNITION

- Featured on Shark Tank India Season 1 — Healthcare innovation pitch
- All-India Winner — Mahindra SAE-India BAJA ATV Competition 2018
- Best Project Award 2018 — Government College of Technology (Maximum Power Point Tracking using Image Processing)
- Certifications: EV Modelling & Simulations; Innovation Management; Financial Modelling & DCF Valuation; Business Branding & Strategy
- Technical Speaker — Electronics For You (EFY) Group
- Member — Society of Automotive Engineers (SAE); National Society of Professional Engineers (NSPE); International Association of Engineers